Do Edits That Fool One Face Recognition Model Fool Others? Measuring the Transfer of PGD Perturbations on LFW

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***Abstract*—Photo-cloaking tools assume that an edit built against one face recognition model will also fool models its author never saw. We test that assumption on the 3,000 same-person pairs of LFW View 2. We trained an InceptionResnetV1 surrogate from scratch on CASIA-WebFace, edited one photo in each pair with L_∞ projected gradient descent (PGD) against it, and scored the edits on four models that played no part in building them: two pretrained InceptionResnetV1 networks, an ArcFace ResNet-50 and an LBPH matcher. At $\varepsilon = 4/255$ (SSIM 0.955) the edit stops the surrogate matching 100.0% of the pairs, but no other deep model loses more than 4.8%. Repeating the attack with each pretrained network as the attacker shows that our surrogate is the weakest source of transferable edits: at $\varepsilon = 8/255$, edits built against the VGGFace2 model break 82.2% of the CASIA model's pairs, against 16.2% for our surrogate. Edits transfer well between networks that share an architecture, even when their training data differ, while ArcFace resists every attacker (at most 18.8%, at a clearly visible $\varepsilon = 16/255$). Random noise, JPEG compression and blur barely affect the deep models. Across three random seeds, no result has a standard deviation above 0.69 percentage points.**

***Index Terms*—adversarial examples, transferability, face recognition, privacy, projected gradient descent**

I. Introduction

Fawkes [1] and LowKey [2] protect people from face recognition by adding small perturbations to photos before they are posted. A photo is edited once, against whatever model the tool's designers had, and must then defeat every system that sees it later. Radiya-Dixit *et al.* [3] point out how weak that position is: the defender commits to an edit, and the recogniser can be replaced or retrained afterwards.

Both tools change the photos a recogniser later learns from or searches. We ask a simpler question about a recogniser that is already trained. If a photo is edited until one model no longer recognises its owner, do models the editor never had access to also stop recognising them? And where some models resist while others give way, does the difference follow their architecture or the faces they were trained on?

We answer with a controlled transfer experiment on LFW [9]. The four test models were chosen so that each differs from our surrogate in a known way. The *casia* model matches its architecture, objective and training data, so only the random initialisation differs; *vggface2* keeps the architecture but saw different faces; ArcFace changes all of these. A hand-crafted LBPH matcher [16] adds a baseline without deep learning. Our contributions are:

- a measurement of how often PGD edits built against a from-scratch surrogate stop four unseen models from matching same-person pairs, over seven edit budgets, with random-noise, JPEG and blur baselines;
- an attacker–target matrix in which every differentiable model takes a turn as the attacker, showing that shared architecture predicts transfer better than shared training data;
- a pipeline in which every model crops the same edited photo its own way, with each model's clean accuracy checked against its published figure before any attack.

II. Related Work

Small, targeted input changes can flip the output of a deep network [4], and gradient-sign methods [5] and PGD [6] find them efficiently. Such perturbations often transfer between models trained for the same task [7], which makes black-box attacks practical. Liu *et al.* [8] found that targeted perturbations rarely transfer from one model to another, and that optimising against an ensemble of models makes them transfer much more often.

Face recognition models map an aligned face crop to an embedding and compare embeddings by distance. FaceNet [12] trained such embeddings with a triplet loss; ArcFace [14] adds an angular margin to a softmax loss. Widely used pretrained models include InceptionResnetV1 [13] networks trained on VGGFace2 [11] and CASIA-WebFace [10], distributed with facenet-pytorch [19], and InsightFace's ArcFace ResNet-50 trained on WebFace600K, a subset of WebFace260M [15]. LFW remains the standard verification benchmark, although it is too small to train on [20] and its subjects are mostly male and White [21].

Fawkes and LowKey use adversarial perturbations as a privacy defence, and both depend on transfer from the model they optimise against to the one a third party runs. Radiya-Dixit *et al.* show that recognisers trained later can learn to ignore such perturbations. We isolate the inference-time question and vary architecture and training data separately.

III. Method

A. Threat model

A user edits one photo of themselves before posting it. They do not know which other photo it will later be compared with, so the attack pushes the edited photo away from the embedding

TABLE I
MODELS. "5-PT" IS FIVE-LANDMARK ALIGNMENT; "BOX" IS AN MTCNN BOX CROP WITH A 32-PIXEL MARGIN.

Model	Architecture	Training data	Loss	Input
surrogate	InceptionResnetV1	CASIA-WebFace (ours)	softmax	5-pt, 160^2
casia	InceptionResnetV1	CASIA-WebFace	softmax	box, 160^2
vggface2	InceptionResnetV1	VGGFace2	softmax	box, 160^2
arcface	ResNet-50	WebFace600K	ArcFace	5-pt, 112^2
lbph	LBP histograms	none	none	5-pt, 112^2

TABLE II
CLEAN LFW VIEW 2 ACCURACY (MEAN ± STD OVER 10 FOLDS) AGAINST PUBLISHED FIGURES.

Model	Ours	Published	Gap (pts)
surrogate	98.12% ± 0.63%	–	–
casia	98.73% ± 0.64%	99.05%	-0.32
vggface2	99.37% ± 0.33%	99.65%	-0.28
arcface	99.85% ± 0.17%	99.83%	+0.02
lbph	67.03% ± 1.03%	–	–

of its own original. The edit succeeds against a recogniser if that recogniser no longer matches the edited photo with a different, unedited photo of the same person.

B. Models

Table I lists the five models. The surrogate is an InceptionResnetV1 that we trained from scratch with a softmax cross-entropy loss on CASIA-WebFace, the same architecture and objective as the two facenet-pytorch models. The *casia* model therefore differs from it only in initialisation, *vggface2* in training data, and *arcface* in architecture, objective and data. LBPH compares an 8×8 grid of uniform local binary pattern histograms with the chi-square distance.

C. Preprocessing

Each model expects faces cropped and aligned its own way. A wrong crop raises no error; it only lowers accuracy, which is easy to miss. We run MTCNN [17] once on every LFW photo (it found a face in all 7,701) and express each model's crop as a 2×3 affine map from the 250×250 photo to the model input: a box crop at 160×160 for the facenet models, the standard InsightFace five-point alignment at 112×112 for ArcFace and LBPH, and a five-point alignment at 160×160 for the surrogate, using the landmark positions MTCNN finds on its CASIA training crops. The crops are applied with a differentiable bilinear warp, so the surrogate's own cropping sits inside the attack. Every edit is made on the full photo, and every model crops the same edited photo.

D. Verification protocol

A pair is accepted as the same person when the distance between embeddings (cosine for the deep models, chi-square for LBPH) is below a threshold. Following the LFW protocol, each model's threshold for a fold is fitted on the clean pairs of the other nine folds. We keep these clean thresholds fixed when scoring edited photos, so $\varepsilon = 0$ gives each model's ordinary error rate.

E. Attack

For a photo x with surrogate embedding $f(x)$, PGD [6] starts from x plus uniform noise in $[-\varepsilon, \varepsilon]$ and repeats

$$x^{t+1} = \Pi_\varepsilon\big(x^t + \alpha \operatorname{sign} \nabla_x \, d\big(f(x^t), f(x)\big)\big)\,, \qquad (1)$$

where d is the cosine distance and Π_ε projects onto the L_∞ ball of radius ε around x and onto valid pixel values. We use 20 steps with $\alpha = 2.5\varepsilon/20$, and round the result to 8 bits before scoring, as a posted photo would be. We sweep $\varepsilon \in \{0, 0.5, 1, 2, 4, 8, 16\}/255$.

F. Baselines and metrics

Random sign noise of size ε on every pixel tests whether the optimisation matters. Saving the edited photo as JPEG at quality 75 tests whether ordinary recompression removes the edit. JPEG compression (quality 90 to 30) and Gaussian blur (σ = 0.5 to 3 pixels) of the clean photo are edits any user could make for free.

Our measure is the *unmatched share*: the fraction of the 3,000 same-person pairs that a model no longer accepts after photo a of each pair is edited, averaged over the ten folds. SSIM [18] between the edited and original photo measures how visible the edit is.

IV. EXPERIMENTAL SETUP

LFW View 2 holds 6,000 fixed pairs in ten folds. We use them unchanged, including the six pairs its maintainers list as mislabelled. The surrogate was trained for 30 epochs on CASIA-WebFace (490,623 aligned images of 10,572 people) with SGD (Nesterov momentum 0.9, weight decay 5×10^{-4}), a one-cycle schedule peaking at a learning rate of 0.1, batch size 256, bf16 mixed precision and random horizontal flips. Training took about 1.5 hours on one RTX 5070 Ti. The final epoch reached 98.12% clean LFW accuracy (best 98.32%, at epoch 27), against 99.05% published for the pretrained *casia* model trained on the same data.

Because a misaligned crop lowers accuracy without any error, we checked every model against its published LFW accuracy before attacking (Table II). We chose the facenet crop margin from $\{0, 14, 32, 44\}$ pixels by clean accuracy; 32 was best for both models. The three pretrained deep models land within 0.32 points of their published figures. For the cross-attacker experiments we converted ArcFace from ONNX to PyTorch with onnx2torch so that gradients flow through it; the converted model reproduces the original's accuracy.

V. RESULTS

A. Transfer from the surrogate

Table III and Fig. 1 give the main result. The surrogate stops matching quickly: from 2.6% of pairs with no edit to 84.8% at $\varepsilon = 2/255$, where the edit is essentially invisible (SSIM 0.987), and 100.0% from $4/255$ up. The other deep models hardly move at these budgets. At $4/255$ *casia* goes from 1.8% to 4.8%, *vggface2* from 0.8% to 1.4% and ArcFace from 0.3%

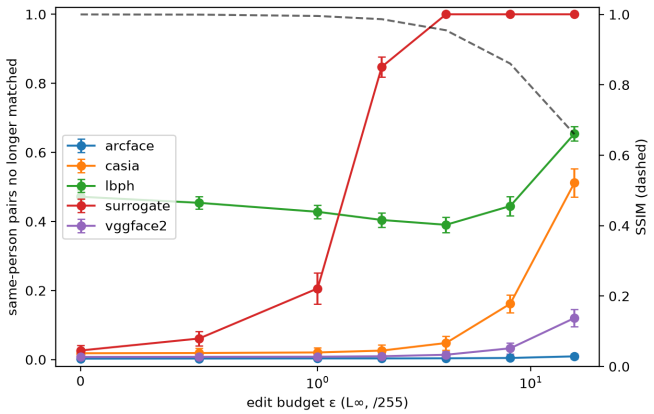


Fig. 1. Unmatched share against edit budget for PGD against the surrogate, one line per model (error bars: std over folds). Dashed: SSIM (right axis).

to 0.4%. Substantial transfer appears only at $16/255$, where SSIM has fallen to 0.660 and the edit is plain to see. There *casia* loses 51.2% of pairs, *vggface2* 12.0% and ArcFace 0.9%.

With our surrogate as the attacker, shared training data looks decisive: *casia*, trained on the same faces, loses 4.3 times as many pairs at $16/255$ as *vggface2*. Section V-D shows that this reading does not survive a change of attacker.

B. Baselines

Random noise does little even though it is more visible than PGD at the same budget (Table IV, Fig. 2). At $\varepsilon = 16/255$ (SSIM 0.462) it moves the surrogate to only 7.4% and *casia* to 4.3%. JPEG at quality 75 after editing weakens small edits, cutting the surrogate's unmatched share at $2/255$ from 84.8% to 39.1%, but edits of $4/255$ and above survive it (97.1% at $4/255$). Free edits barely register: JPEG at quality 30 leaves the surrogate at 3.4%, and only strong blur ($\sigma = 3$ pixels, SSIM 0.790) has an effect, 22.2% for the surrogate and 16.6% for *casia*.

C. Stability across seeds

The random seed sets PGD's starting point and the noise baseline. We repeated the full surrogate sweep with two more seeds. Across the three seeds the standard deviation of every averaged result stayed at or below 0.69 percentage points, far smaller than the differences between models.

D. Every deep model as the attacker

Low transfer could be a property of our surrogate rather than of transfer in general. To test this we repeated the attack with *casia*, *vggface2* and ArcFace as the attacker and scored all five models (Table V, Fig. 3). Edits are equally visible whichever model builds them: at every budget their SSIM differs by at most 0.005.

Our surrogate is the weakest attacker. At $\varepsilon = 8/255$ its edits break 16.2% of *casia*'s pairs, while edits built against *vggface2* break 82.2%. It is also the easiest model to fool: edits from *casia* and *vggface2* break 84.3% and 76.5% of its pairs at the same budget.

Architecture predicts transfer better than training data. The three InceptionResnetV1 models fool one another even when trained on different faces: *vggface2* $\rightarrow$ *casia* reaches 82.2% at $8/255$ and 98.7% at $16/255$. ArcFace resists every attacker; the strongest, *vggface2*, reaches 18.8% and only at $16/255$. ArcFace's own edits transfer moderately to the others (61.9% surrogate, 45.9% *casia*, 21.4% *vggface2* at $16/255$).

Transfer is also lopsided. At $8/255$, *vggface2* $\rightarrow$ *casia* (82.2%) is much stronger than *casia* $\rightarrow$ *vggface2* (32.0%). At the nearly invisible $4/255$, the strongest transfer between two different deep models is *casia* $\rightarrow$ *surrogate* at 30.9%, and at $2/255$ none exceeds 6.1%.

E. What the edits look like

Fig. 4 shows one LFW pair (Vanessa Incontrada) edited by each attacker. At 2–4/255 the edits are hard to see; at $16/255$ they add a visible grain, mostly on skin. Fig. 5 shows the change alone. The pretrained attackers' edits follow the eyes, brows, mouth and outline of the face, while the surrogate's looks like texture with little facial structure. Each edit stays inside its attacker's crop; outside it, only PGD's random start remains.

VI. Discussion

Our first experiment, with only our surrogate as the attacker, suggested that edits transfer mainly to models that share training data. The cross-attacker results overturn that reading. Edits transfer readily among the three InceptionResnetV1 models whatever their training data, and ArcFace, the one model with a different architecture, is barely touched by any of them. On the proposal's question of architecture against data, the evidence favours architecture.

We did not test why our surrogate is a weak attacker, but three explanations fit what we saw. It may overfit: training accuracy reached 97.4% while LFW accuracy levelled off, and we kept the final epoch rather than the best. Its crop geometry differs from the facenet models', which use a box crop; ArcFace, which shares the landmark-based alignment, transfers better to the surrogate (29.6%) than to *casia* (16.2%) at $8/255$. It also had far less training than the pretrained models.

LBPH needs care. With a clean accuracy of 67.0% it already misses 47.2% of same-person pairs before any edit. Random noise at $16/255$ breaks 90.1% of its pairs against 65.4% for PGD, and blur at $\sigma = 3$ breaks 99.6%. It reacts to any change in fine texture, so we treat it as a baseline rather than as evidence of transfer.

This is discouraging for privacy tools. Whether an edit protects someone depends on which model it was built against, and the protection is much weaker against a model with a different architecture. At a nearly invisible budget, even the best-matched pair of models transfers to only 30.9% of pairs.

VII. Limitations

We used one attack and did not try ensemble attacks [8], which usually transfer better. Only the surrogate attacker was

TABLE III

UNMATCHED SHARE OF SAME-PERSON PAIRS (MEAN ± STD OVER FOLDS) AFTER PGD AGAINST THE SURROGATE. $\varepsilon = 0$ IS EACH MODEL'S BASELINE ERROR. SSIM IS BETWEEN EDITED AND ORIGINAL PHOTO.

ε	surrogate	casia	vggface2	arcface	lbph	SSIM
0/255	2.6% ± 1.5%	1.8% ± 1.4%	0.8% ± 0.7%	0.3% ± 0.4%	47.2% ± 2.1%	1.000
0.5/255	6.1% ± 2.1%	1.9% ± 1.4%	0.8% ± 0.7%	0.3% ± 0.4%	45.4% ± 1.8%	0.999
1/255	20.6% ± 4.5%	2.1% ± 1.4%	0.8% ± 0.7%	0.3% ± 0.4%	42.8% ± 1.9%	0.996
2/255	84.8% ± 2.9%	2.6% ± 1.7%	1.0% ± 0.8%	0.3% ± 0.4%	40.4% ± 2.1%	0.987
4/255	100.0% ± 0.0%	4.8% ± 2.0%	1.4% ± 0.9%	0.4% ± 0.5%	39.0% ± 2.2%	0.955
8/255	100.0% ± 0.0%	16.2% ± 2.6%	3.3% ± 1.6%	0.5% ± 0.5%	44.5% ± 2.8%	0.860
16/255	100.0% ± 0.0%	51.2% ± 4.1%	12.0% ± 2.6%	0.9% ± 0.7%	65.4% ± 2.1%	0.660

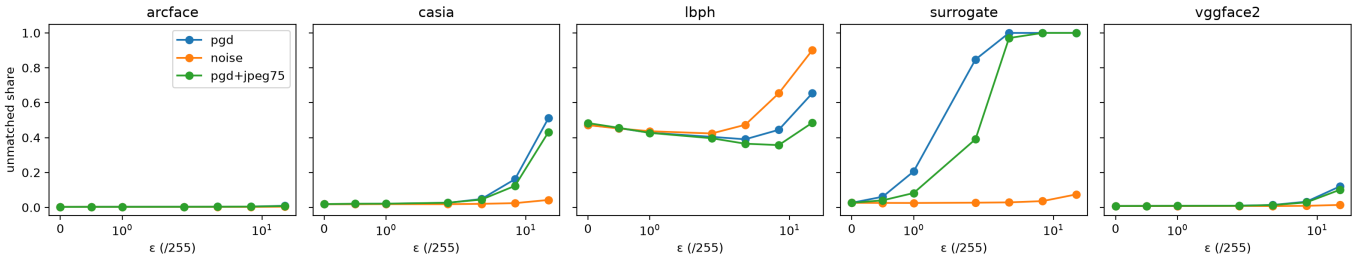


Fig. 2. PGD against the surrogate compared with random noise of the same budget and with PGD followed by JPEG at quality 75, one panel per scored model.

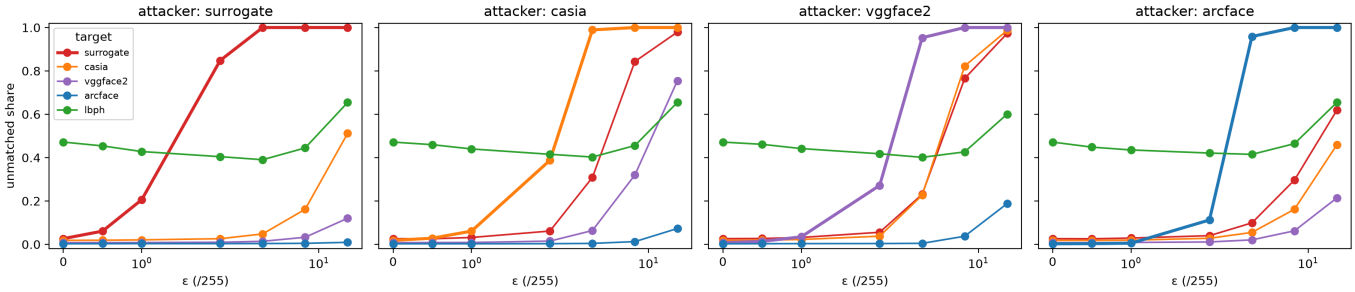


Fig. 3. Unmatched share against ε for each attacker (one panel each), one line per target. The thick line is the attacker scored on itself.

TABLE IV

UNMATCHED SHARE UNDER BASELINE EDITS (S SURROGATE, C CASIA, V VGGFACE2, A ARCFACE, L LBPH).

Edit	Strength	S	C	V	A	L	SSIM
Noise	ε = 8/255	3.6%	2.4%	0.9%	0.3%	65.4%	0.713
Noise	ε = 16/255	7.4%	4.3%	1.4%	0.4%	90.1%	0.462
PGD + JPEG 75	ε = 2/255	39.1%	2.7%	0.9%	0.3%	39.6%	0.968
PGD + JPEG 75	ε = 4/255	97.1%	4.5%	1.3%	0.4%	36.5%	0.960
PGD + JPEG 75	ε = 16/255	100.0%	43.2%	10.1%	0.8%	48.5%	0.820
JPEG	q = 50	2.8%	1.9%	0.8%	0.3%	49.2%	0.952
JPEG	q = 30	3.4%	2.0%	0.8%	0.3%	54.9%	0.927
Blur	σ = 2	7.9%	5.2%	2.1%	0.3%	97.4%	0.876
Blur	σ = 3	22.2%	16.6%	6.8%	0.7%	99.6%	0.790

TABLE V

UNMATCHED SHARE WHEN THE EDIT IS BUILT AGAINST THE ATTACKER (ROW) AND SCORED ON THE TARGET (COLUMN). BOLD: ATTACKER SCORED ON ITSELF.

Attacker ↓ / target →	surrogate	casia	vggface2	arcface	lbph	SSIM
ε = 4/255						
surrogate	**100.0%**	4.8%	1.4%	0.4%	39.0%	0.955
casia	30.9%	**98.9%**	6.4%	0.4%	40.3%	0.953
vggface2	23.1%	22.7%	**95.3%**	0.5%	40.1%	0.954
arcface	9.9%	5.5%	2.1%	**95.9%**	41.5%	0.953
ε = 8/255						
surrogate	**100.0%**	16.2%	3.3%	0.5%	44.5%	0.860
casia	84.3%	**100.0%**	32.0%	1.3%	45.6%	0.856
vggface2	76.5%	82.2%	**100.0%**	3.7%	42.6%	0.857
arcface	29.6%	16.2%	6.2%	**100.0%**	46.4%	0.856
ε = 16/255						
surrogate	**100.0%**	51.2%	12.0%	0.9%	65.4%	0.660
casia	97.9%	**100.0%**	75.5%	7.3%	65.5%	0.656
vggface2	97.3%	98.7%	**100.0%**	18.8%	60.1%	0.657
arcface	61.9%	45.9%	21.4%	**100.0%**	65.5%	0.655

repeated over three seeds. Landmarks are detected on the clean photo and reused for its edited versions, whereas a deployed system would detect the face again. PGD on the GPU is not bit-for-bit reproducible even with a fixed seed; re-running the example edits changed 3 of 300 fooled-or-matched decisions, all for models sitting at their threshold. PGD's random start leaves noise outside the attacker's crop, which lowers SSIM slightly and equally for every attacker. Finally,

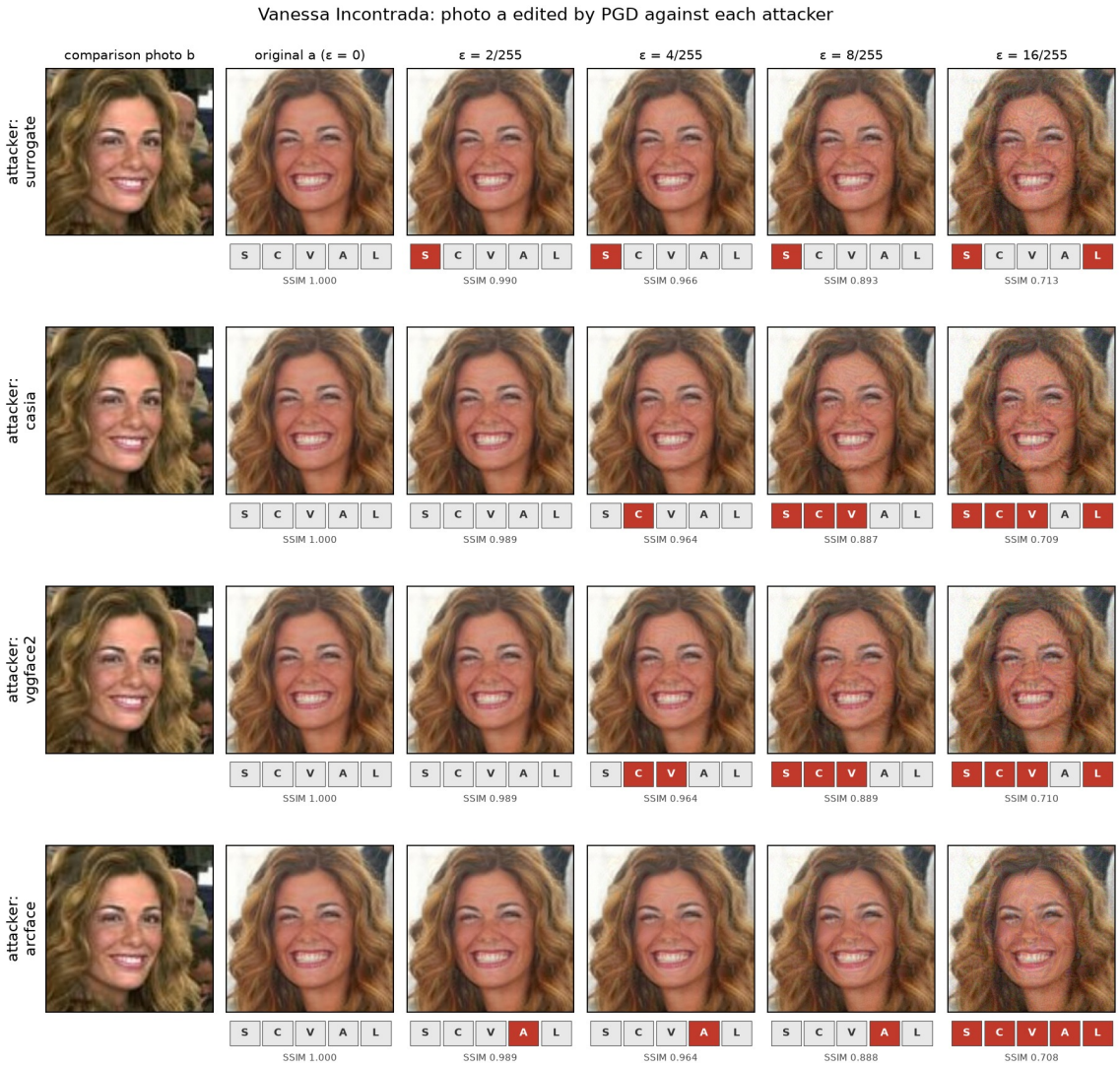


Fig. 4. Photo a of one pair edited against each attacker (rows) at each budget (columns), with the comparison photo b on the left. Tags under each photo: models fooled (red) or still matching (grey); S surrogate, C casia, V vggface2, A arcface, L lbph.

we did not check whether protection is equally effective across demographic groups, which matters given LFW's skew [21].

VIII. Conclusion

PGD edits built against a from-scratch face recognition model transfer poorly to models its author never used. Changing the attacker shows that this is partly a weakness of our surrogate: edits built against a pretrained InceptionResnetV1 transfer strongly to other networks with that architecture, even across training sets. A different architecture resists edits from every attacker we tried until the edit is clearly visible. A privacy edit made once against one model should not be expected to protect against an unknown recogniser.

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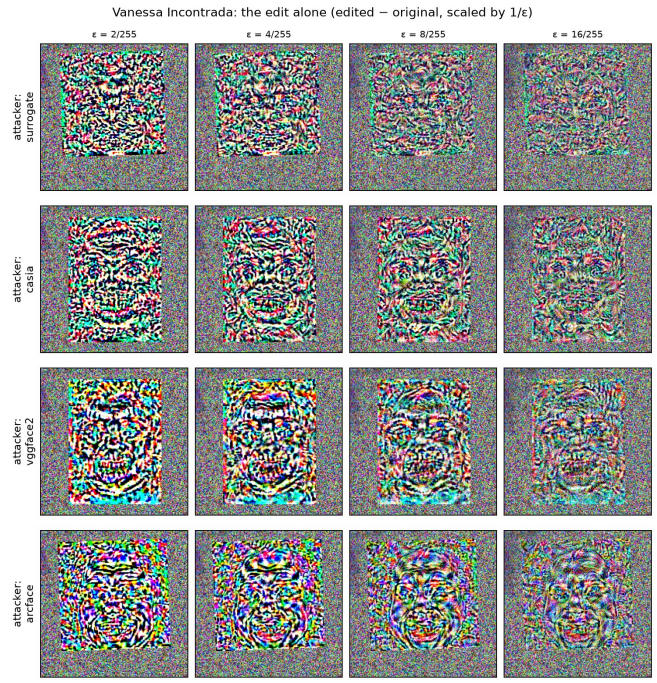


Fig. 5. The edit alone (edited minus original) for the pair in Fig. 4, each column scaled by $1/\varepsilon$.

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